

Homework 12

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Problem 9.2

Let X be an exponential random variable with parameter $\lambda = \frac{1}{2}$.

- (a) Use Markov's inequality to find an upper bound for $P(X > 6)$.
- (b) Use Chebyshev's inequality to find an upper bound for $P(X > 6)$.
- (c) Explicitly compute the probability above and compare with the upper bounds you derived.

Solution.

Scratch / Setup. For $X \sim \text{Exp}(\lambda)$ with $\lambda = \frac{1}{2}$,

$$E[X] = \frac{1}{\lambda} = 2, \quad \text{Var}(X) = \frac{1}{\lambda^2} = 4.$$

Relevant facts:

- $X \geq 0$ almost surely, so Markov's inequality applies.
- Chebyshev's inequality says

$$P(|X - E[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

- Since $E[X] = 2$, the event $\{X > 6\}$ implies $\{|X - 2| \geq 4\}$.
- For an exponential random variable,

$$P(X > x) = e^{-\lambda x}, \quad x \geq 0.$$

- (a) Since $X \geq 0$, Markov's inequality gives

$$P(X > 6) \leq \frac{E[X]}{6} = \frac{2}{6} = \frac{1}{3}.$$

Final: $P(X > 6) \leq \frac{1}{3}$

- (b) Because $\{X > 6\} \subseteq \{|X - 2| \geq 4\}$, we have

$$P(X > 6) \leq P(|X - 2| \geq 4).$$

Now apply Chebyshev's inequality:

$$P(|X - 2| \geq 4) \leq \frac{\text{Var}(X)}{4^2} = \frac{4}{16} = \frac{1}{4}.$$

Therefore

$$P(X > 6) \leq \frac{1}{4}.$$

Final: $P(X > 6) \leq \frac{1}{4}$

(c) The exponential tail can be computed exactly:

$$P(X > 6) = e^{-\lambda \cdot 6} = e^{-3} \approx 0.0498.$$

Comparing the three values,

$$e^{-3} \approx 0.0498 < \frac{1}{4} < \frac{1}{3}.$$

So both inequalities give valid upper bounds, and the Chebyshev bound is better than the Markov bound because $\frac{1}{4}$ is smaller and closer to the exact probability.

Final: $P(X > 6) = e^{-3} \approx 0.0498,$ and Chebyshev gives the better upper bound

Problem 9.6

Nate is a competitive eater specializing in eating hot dogs. From past experience we know that it takes him on average 15 seconds to consume one hot dog, with a standard deviation of 4 seconds. In this year's hot dog eating contest he hopes to consume 64 hot dogs in just 15 minutes. Use the CLT to approximate the probability that he achieves this feat of skill.

Solution.

Scratch / Setup. Let X_i be the time, in seconds, that Nate needs to consume the i th hot dog, and let

$$S_{64} = X_1 + \cdots + X_{64}.$$

Then Nate succeeds exactly when $S_{64} \leq 15 \text{ minutes} = 900 \text{ seconds}$.

From the data,

$$E[X_i] = 15, \quad \text{SD}(X_i) = 4, \quad \text{Var}(X_i) = 16.$$

Assuming the X_i are i.i.d.,

$$E[S_{64}] = 64 \cdot 15 = 960, \quad \text{Var}(S_{64}) = 64 \cdot 16 = 1024, \quad \text{SD}(S_{64}) = 32.$$

Use the CLT because the exact distribution of S_{64} is not given:

$$\frac{S_{64} - 960}{32} \approx N(0, 1).$$

We want the probability that Nate consumes 64 hot dogs within 15 minutes, so

$$P(\text{Nate succeeds}) = P(S_{64} \leq 900).$$

By the central limit theorem,

$$P(S_{64} \leq 900) \approx P\left(\frac{S_{64} - 960}{32} \leq \frac{900 - 960}{32}\right) = \Phi\left(\frac{-60}{32}\right) = \Phi(-1.875).$$

Therefore,

$$P(S_{64} \leq 900) \approx \Phi(-1.875) \approx 0.0304.$$

So the CLT approximation says that Nate has about a 3.04% chance of finishing 64 hot dogs in 15 minutes.

Final: $P(S_{64} \leq 900) \approx \Phi(-1.875) \approx 0.0304$

Problem 9.18

Suppose that $X \sim \text{NegBin}(100, \frac{1}{3})$.

- (a) Give an upper bound for $P(X > 500)$ using Markov's inequality.
- (b) Give an upper bound for $P(X > 500)$ using Chebyshev's inequality.
- (c) Recall that X can be written as the sum of 100 i.i.d. random variables with $\text{Geom}(\frac{1}{3})$ distribution. Use this representation and the CLT to estimate $P(X > 500)$.
- (d) Recall that X can be represented as the number of trials needed to get the 100th success in a sequence of i.i.d. trials of success probability $\frac{1}{3}$. Estimate $P(X > 500)$ by looking at the distribution of the number of successes within the first 500 trials.

Solution.

Scratch / Setup. Here $X \sim \text{NegBin}(100, \frac{1}{3})$ means that X is the number of trials needed to obtain the 100th success when each trial succeeds with probability

$$p = \frac{1}{3}.$$

With $r = 100$ and this convention,

$$E[X] = \frac{r}{p} = \frac{100}{1/3} = 300,$$

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{100(2/3)}{(1/3)^2} = 600, \quad \text{SD}(X) = \sqrt{600} = 10\sqrt{6}.$$

For part (c), write

$$X = G_1 + \cdots + G_{100},$$

where the G_i are i.i.d. $\text{Geom}(\frac{1}{3})$ random variables.

For part (d), let N be the number of successes in the first 500 trials. Then

$$N \sim \text{Bin}\left(500, \frac{1}{3}\right), \quad \{X > 500\} = \{N \leq 99\}.$$

Also,

$$E[N] = \frac{500}{3}, \quad \text{Var}(N) = 500 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{1000}{9}, \quad \text{SD}(N) = \sqrt{\frac{1000}{9}}.$$

- (a) Since $X \geq 0$, Markov's inequality applies:

$$P(X > 500) \leq \frac{E[X]}{500} = \frac{300}{500} = \frac{3}{5}.$$

Final: $\boxed{P(X > 500) \leq \frac{3}{5}}$

(b) Since $X > 500$ implies $X - 300 > 200$, we have

$$\{X > 500\} \subseteq \{|X - 300| \geq 200\}.$$

Hence Chebyshev's inequality gives

$$P(X > 500) \leq P(|X - 300| \geq 200) \leq \frac{\text{Var}(X)}{200^2} = \frac{600}{40000} = \frac{3}{200} = 0.015.$$

Final: $P(X > 500) \leq \frac{3}{200} = 0.015$

(c) Using the representation $X = G_1 + \cdots + G_{100}$ and the CLT,

$$\frac{X - 300}{\sqrt{600}} \approx N(0, 1).$$

Therefore

$$P(X > 500) = P\left(\frac{X - 300}{\sqrt{600}} > \frac{500 - 300}{\sqrt{600}}\right) \approx 1 - \Phi\left(\frac{200}{\sqrt{600}}\right).$$

Since

$$\frac{200}{\sqrt{600}} = \frac{20}{\sqrt{6}} \approx 8.165,$$

we get

$$P(X > 500) \approx 1 - \Phi\left(\frac{20}{\sqrt{6}}\right) \approx 1.6 \times 10^{-16}.$$

Final: $P(X > 500) \approx 1 - \Phi\left(\frac{20}{\sqrt{6}}\right) \approx 1.6 \times 10^{-16}$

(d) Let N be the number of successes in the first 500 trials. Then

$$N \sim \text{Bin}\left(500, \frac{1}{3}\right)$$

and

$$P(X > 500) = P(N \leq 99).$$

Using the CLT with a continuity correction,

$$P(N \leq 99) \approx P(N < 99.5) = P\left(\frac{N - 500/3}{\sqrt{1000/9}} < \frac{99.5 - 500/3}{\sqrt{1000/9}}\right) \approx \Phi\left(\frac{99.5 - 500/3}{\sqrt{1000/9}}\right).$$

Now

$$\frac{99.5 - 500/3}{\sqrt{1000/9}} \approx -6.372,$$

so

$$P(X > 500) = P(N \leq 99) \approx \Phi(-6.372) \approx 9.3 \times 10^{-11}.$$

This is different from part (c), which is not surprising: both are CLT approximations, and this is an extreme tail event.

Final: $P(X > 500) = P(N \leq 99) \approx \Phi(-6.372) \approx 9.3 \times 10^{-11}$